

$$P(F_3|E) = \frac{P(E|F_3)P(F_3)}{\sum_i P(E|F_i)P(F_i)} \sim \text{BAYES + TOTAL P}$$

EXAMPLE

LAB TEST < 99% ACCURATE $P(P|D)$
 1% FALSE POSITIVE $P(P|D^c)$

~~50%~~

$P(D)$
 0.5% OF POPULATION ARE ILL (PREVALENCE)

IF ONLY COME OUT POSITIVE, ON WHAT % IS HE ILL?

OUTCOME: (TEST RESULT ILL OR NOT)

$$P(D|P) = \frac{P(P|D) \cdot P(D)}{P(P|D)P(D) + P(P|D^c)P(D^c)}$$

YOU HAVE DISEASE GIVEN THAT YOU TEST POSITIVE

$\approx \frac{1}{3}$

IS IT STILL USEFUL? YES

$P(D) = 0.5\%$ BAYESIAN PRIOR

→ IF THIS ONE IS LOW THEN A SMALL FALSE POSITIVE WILL HAVE BIG EFFECT

INDEPENDENT E.

$$P(E|F) = P(E)P(F)$$

$$P(E|F) = \frac{P(E|F)}{P(F)} = \frac{P(E)P(F)}{P(F)} = P(E)$$

PROPOSITIONS:

- 1) $E \text{ IND } F \Rightarrow E \text{ IND } F^c$
- 2) $E \text{ IND } F \Rightarrow F \text{ IND } E$

3) $\underbrace{U \text{ IND } A}_{\text{IND } G}$ ~~$P \text{ IND } G$~~

INDEPENDENCE IS "ASSUMED"

RELATED THINGS

M TIMES INDEPENDENT CONDITIONS

$\left(\frac{H}{T}\right)$

M TIMES COMPUTER SIMULATION

~~~~~

(GIVEN THAT CACHE IS EMPTY ETC.)



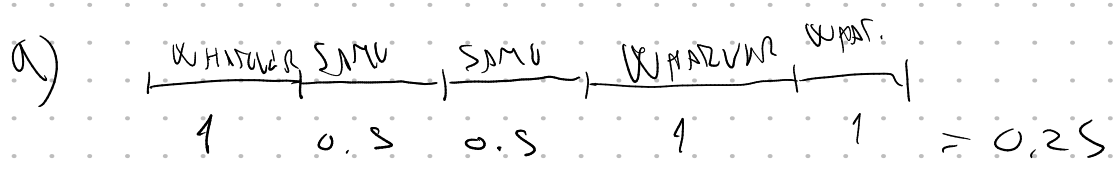
Coin FLIPPED 5 TIMES

a) 1<sup>st</sup> 3 FLIP SAME OUTCOME

b) 1<sup>st</sup> 3 OR LAST 3 SAME OUTCOME

c) AT LEAST 2 H IN 1<sup>st</sup> 3 FLIPS AND 2 T IN LAST 3

WE NEED THE COIN TO BE FAIR TO USE THE UNIFORM P. MODEL



USING UPM

32 POSSIBLE O.

2 FAVOURABLE

$$\begin{array}{cc}
 H & H & H & X & X \\
 T & T & T & X & X
 \end{array}
 \rightsquigarrow 2 \cdot 2^2 = 8 \Rightarrow \frac{8}{32}$$

b)

F | 1<sup>st</sup> 3 SAME OUTCOME

L | LAST 3 SAME

$P(F \cup L) = P(F) + P(L) - P(FL)$

$P(F) = P(L)$ ,

$P(FL) = P(F)P(L)$

$\uparrow \uparrow = \frac{1}{2} - \frac{1}{16} = \frac{7}{16}$

c) AT LEAST 2 H IN 1<sup>st</sup> 3 FLIPS AND 2 T IN LAST 3

VPM

$$P(E) = \frac{|E|}{|S|} = \frac{6}{32}$$

2 → X H H T T  
 1 H ~~X~~ H T T  
 1 H H ~~X~~ T T  
 1 H H T ~~X~~ T  
 1 H H T T ~~X~~  
 6

ONLY 1 TO NOS  
 DUPLICATED

5 4 3 2 1  
 1 1 1 1 1  
 1 1 1 1 1  
 1 1 1 1 1

EXERCISE

MARIO ROSS, M KEYS  $> 1$

1) CHANGES ONE AT A TIME

2) DISCARDS IF IT'S THE WRONG ONE

2.1) IF HE IS DRUNK, HE DOESN'T DISCARD

a) P OPENS DOOR ON KTH ATTEMPT

b) P . . . . . WITHIN K ATTEMPTS

OUTCOME: SEQUENCE OF KEYS

PERMUTATION OF M ELEMENTS ( $M!$  OUTCOMES)

a)  $\frac{1 \cdot (M-1)!}{M!} = \frac{1}{M}$

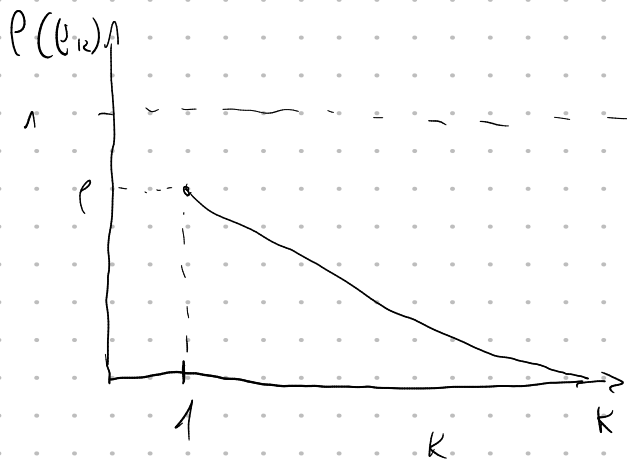
b)  $\frac{K}{M}$

VARIATION: MARIO IS DRUNK, SO HE PUT THE KEY BACK INSTEAD OF TRYING A NEW ONE

NOW EVERY ATTEMPT IS INDEPENDENT

a.1)  $P = \frac{1}{M}$   $1 - P = 1 - \frac{1}{M}$

$P(E_k) = (1 - P)^{k-1} P$



$$\lim_{k \rightarrow +\infty} P(E_k) = 0$$

$$b.1) P\left(\bigcup_{i=1}^k E_i\right) = \sum_{i=1}^k P(E_i) = \sum_{i=1}^k (1-p)^{i-1} \cdot p$$

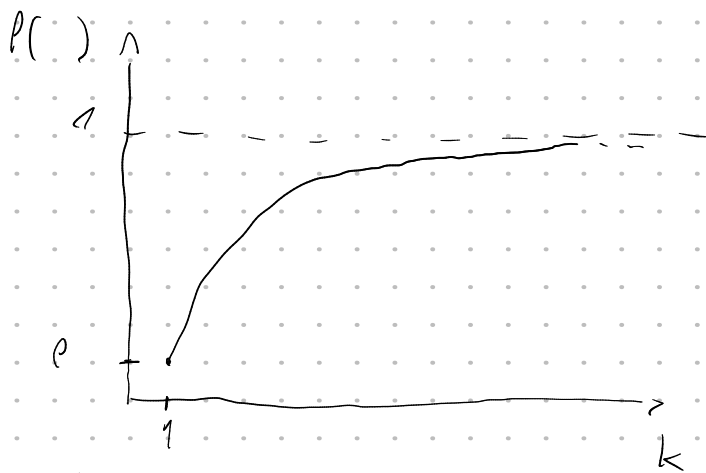
WE CAN USE GEOMETRIC SUM TO GET A CLOSED FORMULA

$$\sum_{i=0}^M x^i = \frac{1 - x^{M+1}}{1 - x}$$

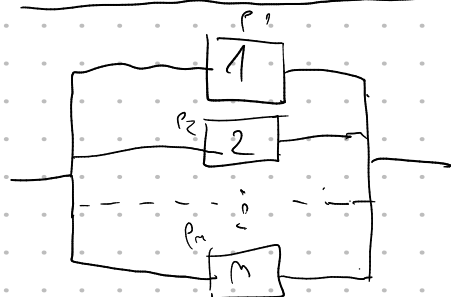
GEOMETRIC SUM

IT'S EASIER TO FIND  $P(b.1^c) \rightarrow$  GOT IT WRONG K TIMES  
 $= (1-p)^k$

$$P\left(\bigcup_{i=1}^k E_i\right) = 1 - (1-p)^k$$



PARALLEL SYSTEMS



SYSTEM WORK IF AT LEAST ONE WORKS

$p_i$ : IT'S SWITCH CLOSED

WE ASSUME THAT PARALLEL COMPONENTS ARE INDEPENDENT

$$P(\text{AT LEAST 1 WORK})$$

$$= 1 - P(\text{ALL BREAK})$$

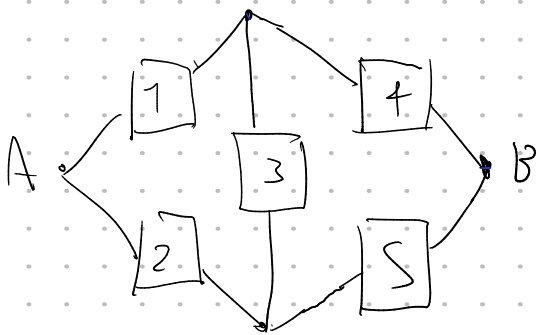
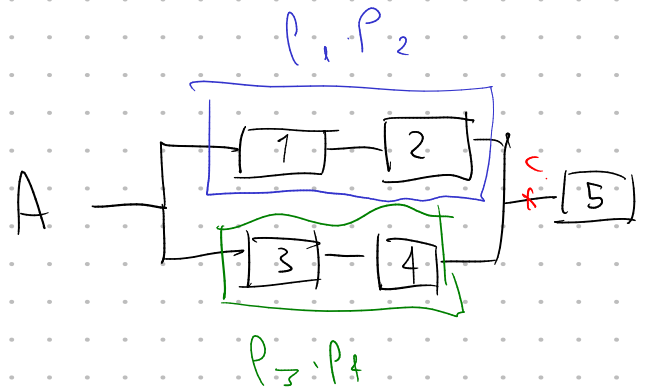
$$= 1 - P(A_1^c A_2^c \dots A_m^c)$$

$$= 1 - \prod_{i=1}^m P(A_i^c)$$

—————○—————○

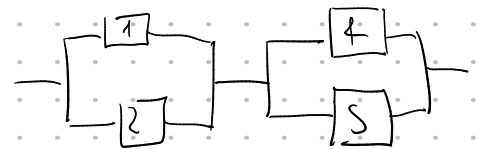
$$P(A \rightarrow B) = P(A \rightarrow C) \cdot P(C \rightarrow B)$$

$$= [1 - (1 - p_1 p_2)(1 - p_3 p_4)] p_5$$



~> WE CAN DO CONDITIONING!

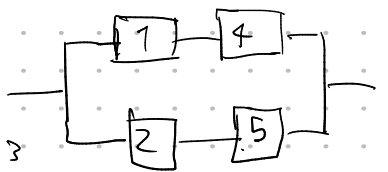
3 CASES



$$[1 - (1 - p_1 p_4)(1 - p_2 p_5)] (1 - p_3) +$$

$$[1 - (1 - p_1)(1 - p_2)] \cdot [1 - (1 - p_4)(1 - p_5)] p_3$$

3 OTHER



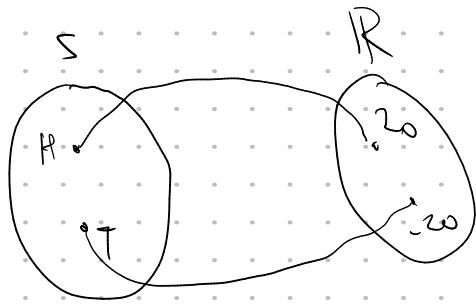
RANDOM VARIABLES

REAL-VALUED FUNCTION

$$X: S \rightarrow \mathbb{R}$$

RANDOM PART IS IN S (CIRCLED)

## EXAMPLE



$$X(H) = +20$$

$$X(T) = -20$$

$$P\{X = +20, -20\} = \frac{1}{2}$$

~~~~~> ABSTRACTION

SUPPORT: SET OF VALUES THAT THE RANDOM VARIABLE CAN TAKE
 $\mathbb{R}$

SUPPORT CAN BE $\begin{cases} \text{CONTINUOUS} & (\text{AN INTERVAL}) \text{ TYPE OF FREQUENCY} \\ \text{DISCRETE} & (\text{A FINITE SET OF NUMBERS}) \end{cases}$

ON THIS WE DEFINE R.V.

X CUMULATIVE DISTRIBUTION FUNCTION (CDF)

$$F_X(w) = P\{X \leq w\}$$

$\mathbb{R}$ LOWER CASE
 $\mathbb{R}$ V UPPER CASE

PROPR.

IT MATTERS THAT IT IS $w \leq x$

- 1) CDF IS RIGHT CONTINUOUS
- 2) WEAKLY MONOTONIC (IT CAN ONLY GROW)
- 3) $\lim_{w \rightarrow +\infty} F(w) = 1$
- 4) $\lim_{w \rightarrow -\infty} F(w) = 0$

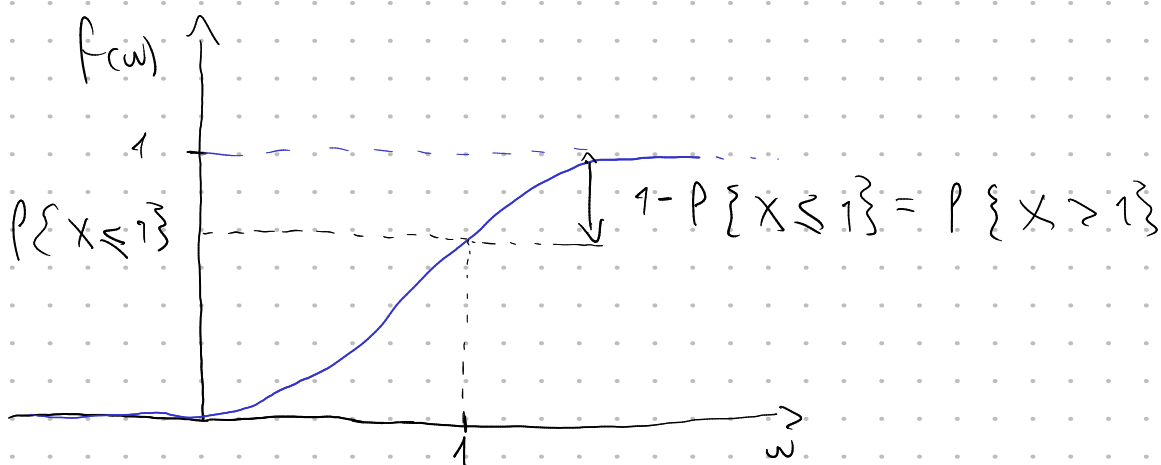
EXERCISE

R.V. x

$$F(w) = \begin{cases} 0 & w < 0 \\ 1 - e^{-w^2} & w \geq 0 \end{cases}$$

$$a) \quad P\{X > 1\}$$

DRAW THE THING FIRST
CONTINUOUS RV



$$a \sim 1 - (1 - e^{-1^2}) = e^{-1} = \frac{1}{e}$$

CDF COMPLETELY CHARACTERIZES THE RANDOM VARIABLE
EXAMPLE

$$P\{1 < X \leq 2\} = ?$$

$$P\{X \leq 2\} = P\{X \leq 1\} + P\{1 < X \leq 2\}$$

$$P\{1 < X \leq 2\} = P\{X \leq 2\} - P\{X \leq 1\}$$

PROBABILITY MASS (DISCRETE RV ONLY) PMF

PROBABILITY MASS FUNCTION (PMF)

$$p(a) = P\{X = a\}$$

$$\sum_{-\infty}^{+\infty} p(a) = 1$$

FROM PMF TO CDF

$$F(a) = \sum_{x \leq a} p(x)$$

FROM CDF TO PMF

$$p(a) = F(a) - F(a^-)$$

CONTINUOUS RV ONLY PMF DOESN'T WORK: (NONSENSE)

$$P(X=a) = \emptyset$$

PROBABILITY DENSITY FUNCTION

PDF

$f(x)$ PDF $\forall B \subseteq \mathbb{R}$

$$P\{X \in B\} = \int_B f(x) dx$$

$$\text{If } B = (-\infty, +\infty) \rightarrow P\{X \in (-\infty, +\infty)\} = 1$$

$$\int_{-\infty}^{+\infty} f(x) dx = 1$$

ASSUME

$$B = [a, b]$$

$$P\{a \leq X \leq b\} = \int_a^b f(x) dx$$

||

$$F(b) - F(a) =$$

$$\text{So, if } B \equiv \left[a - \frac{\varepsilon}{2}, a + \frac{\varepsilon}{2} \right]$$

$$P\{X \in [a - \frac{\varepsilon}{2}, a + \frac{\varepsilon}{2}]\} = \int_{a - \frac{\varepsilon}{2}}^{a + \frac{\varepsilon}{2}} f(x) dx \approx f(a) \cdot \varepsilon$$

PDF CDF RELATIONSHIP:

$$F(a) = P\{X \leq a\} = P\{-\infty < X \leq a\}$$

$$= \int_{-\infty}^a f(x) dx$$

CDF PDF

$$f_a = \frac{dF(a)}{da}$$